Performance Enhancement of FTTH Networks through Gain and Power Optimization in WDM-PON Architecture

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***Abstract*—This research focuses on the analysis of gain and power optimization for improved coverage in Fiber-to-the-Home (FTTH) networks using Wavelength Division Multiplexing Passive Optical Network (WDM-PON) architecture. The study evaluates system performance based on optical power budget, splitter ratio, and network reach under various channel conditions. By employing multiple wavelength channels, the WDM-PON system enhances data capacity, minimizes signal degradation, and ensures better service quality for end-users. The simulation results demonstrate that proper amplification and power management significantly improve overall system gain, reduce attenuation losses, and extend the transmission distance while maintaining acceptable Bit Error Rate (BER) and Quality Factor (Q-factor) levels. The proposed design achieves efficient bandwidth utilization and stable transmission performance, making it a promising approach for next-generation broadband access networks and high-speed optical communication systems. A search approach was utilized to gather research papers on the design, performance, and optical amplification techniques of Erbium-Doped Fiber Amplifiers (EDFAs) and Raman amplifiers in real-life FTTh deployment. The investigation demonstrates that the Mach-Zehnder Modulator (MZM) can sustain high signal integrity over medium and long distances without quality degradation. The investigation shows that controlling optical gain and power levels can greatly improve system performance. This study highlights MZM as a valuable component for creating scalable, long-reach networks.**

I. INTRODUCTION

Fiber-to-the-Home (FTTH) is evolving as the most intriguing broadband access technology, delivering high-capacity, ultra-low-latency connectivity to end customers. Wavelength Division Multiplexing Passive Optical Network (WDM-PON) systems offer improved bandwidth efficiency and link dependability compared to traditional Time Division Multiplexing PON (TDM-PON) architectures [1]. WDM-PON allows each ONU to operate at a specific wavelength, boosting scalability, signal isolation, and data throughput without sharing time slots [2].

A key component of WDM-PON design is the evaluation of the optical power budget and gain, which influences the maximum transmission distance, network reach, and service quality in FTTH applications [3]. Optical amplifiers, including Erbium-Doped Fiber Amplifiers (EDFAs), Raman amplifiers, and Semiconductor Optical Amplifiers (SOAs), are crucial for enhancing optical signals while preserving low noise figures and minimal distortion [4]. An analysis of a balanced power budget guarantees effective optical power distribution among several users, preventing nonlinearities and signal loss caused by excessive gain [5].

Recent investigations into high-capacity Wavelength Division Multiplexing Passive Optical Network (WDM-PON) systems have elucidated that the optimization of launch power, splitter ratio, and receiver sensitivity can substantially enhance the Quality Factor (Q-factor) and diminish the Bit Error Rate (BER), thereby facilitating longer transmission distances of up to 50 kilometres for Fiber to the Home (FTTH) implementations [6]. Furthermore, hybrid amplification methodologies that integrate Raman and Erbium-Doped Fiber Amplifier (EDFA) technologies have exhibited exceptional performance in long-reach optical access networks [7]. Specifically, bidirectional transmission methodologies employing adaptive gain control augment both upstream and downstream signal integrity amidst fluctuating load conditions [8].

In this research, a WDM-PON-based FTTH network is designed and tested to assess the gain and power performance using simulation models. Several optical channels were combined across the C- and L-bands, taking advantage of the low-loss transmission range of single-mode optical fibers [9]. The examination of power variation, signal reduction, and interference was conducted across various link lengths and amplifier setups to guarantee consistent coverage and maximum energy efficiency. The results show that the suggested setup provides a consistent optical gain of about 18–22 dB, with bit error rates (BER) lower than 10^{-} and Q-factors greater than 15, guaranteeing strong signal transmission over long distances, making it suitable for crowded FTTH settings [10].

Furthermore, energy-efficient hybrid WDM-TDM PON architectures have been proposed for scalable deployment while maintaining cost efficiency and low power consumption [11]. Simulation results confirm that careful optimization of gain and power leads to significant improvements in network range and stability, making this design a strong candidate for next-generation broadband infrastructure.

The analysis shows that optimizing gain and power in the WDM-PON architecture is a key factor in expanding FTTH network coverage, improving reliability, and achieving high-speed data transmission with minimal error rates. The system can efficiently meet growing bandwidth demands for smart homes, IoT and future 5G/6G backhaul applications.

II. LITERATURE REVIEW

Fiber-to-the-Home (FTTH) networks have surpassed previous copper-based and wireless access systems due to their high bandwidth capacity, low latency, and immunity to electromagnetic interference [12]. However, as the demand for ultra-broadband services develops, conventional TDM-PON systems suffer constraints in capacity and reach, principally due to limited power budgets and the requirement to distribute bandwidth among several users [13]. Wavelength Division Multiplexing Passive Optical Networks (WDM-PONs) enable simultaneous transmission of numerous optical channels over the same fiber by designating unique wavelengths to each user [14].

Recent research indicates that gain optimization and power management are crucial factors in enhancing FTTH coverage, guaranteeing dependable signal transmission, and achieving a balance between energy efficiency and performance [15]. Optimizing the gain of optical amplifiers and reducing insertion losses in splitters and multiplexers allows for extending the transmission distance past 50 km while ensuring acceptable Bit Error Rate (BER) and Q-factor performance [16]. Specifically, Erbium-Doped Fiber Amplifiers (EDFAs) and Raman amplifiers have been widely utilized to improve link performance while minimizing noise or distortion [17].

Despite this, the degradation of optical signals, caused by attenuation, dispersion, and nonlinear effects, remains a major challenge in long-reach Wavelength Division Multiplexing Passive Optical Networks (WDM-PON) architectures [18]. Fiber impairments, including Four-Wave Mixing (FWM), Cross-Phase Modulation (XPM), and Self-Phase Modulation (SPM), can substantially undermine power uniformity and signal integrity across various wavelengths [19]. In order to address these concerns, adaptive power control mechanisms alongside hybrid amplifier configurations have been proposed to uphold consistent gain profiles and mitigate noise accumulation [20]. Moreover, effective power budgeting not only enhances communication in both downstream and upstream directions but also ensures a balanced distribution of optical power among Optical Network Units (ONUs), thereby minimizing system-wide power fluctuations and augmenting user coverage [1].

As stated by Zhang et al. According to [3], multi-band wavelength division multiplexing (WDM) optical systems that function in the C+L bands demonstrate enhanced spectral efficiency and extended range when paired with advanced fiber optics like G. 654, which has lower signal loss. The use of gain flattening filters (GFFs) and automatic gain control (AGC) circuits enhances the stability of optical power levels and addresses variations in gain that depend on wavelength [12]. In a similar manner, research conducted by Christodoulou et al. [13] and De Santi and colleagues. Research by [4] has shown that using adaptive scheduling algorithms and "Super-PON" structures greatly enhances resource distribution, reduces energy waste, and increases the scalability of networks.

Energy-effective deployment models like mongrel WDM-TDM PONs have also been explored to lower capital and functional expenditures while maintaining high content and capacity [20]. By using dynamic gain equalization and optimized splitter rates, WDM-PON systems can achieve invariant performance across large subscriber bases without immolating signal quality or outturn.

In summary, assaying gain and power optimization within the WDM-PON frame is critical for achieving better content, advanced trustability, and sustainable energy effectiveness in FTTH networks. The integration of advanced modification ways, adaptive gain control, and precise power budgeting offers a pathway toward next-generation broadband structure capable of meeting the raising demands of 5G/6G backhaul, IoT, and smart megacity operations.

III. METHODOLOGY

A. *Search Strategy*

A comprehensive search strategy was in place to gather sufficient material for this examination of the Fiber-to-the-Home (FTTH) overlaying Wavelength Division Multiplexing Passive Optical Network (WDM-PON) architecture. Our goal was to compile research articles that examine WDM-PON design, performance metrics, and optical amplification methods for Raman amplifiers and Erbium-Doped Fiber Amplifiers (EDFAs) in a FTTH context. When selecting search phrases, Google Scholar and many reliable academic resources, such as IEEE Xplore, ScienceDirect, and SpringerLink, were examined. To refine findings and make sure the team didn't miss any pertinent publications, boolean searches and phrase manipulations were used. "WDM-PON," "FTTH architecture," "optical amplifier," "EDFA," "Raman amplification," and "long-reach optical network" were among the search phrases used. The Boolean notions "AND" and "OR" were used to link key word combinations, such as "WDM-PON AND FTTH," "FTTH architecture AND optical amplifier," and "EDFA OR Raman amplifier AND WDM-PON."

We only included conference papers and peer-reviewed journal articles published between 2018 and 2025 in order to increase the caliber and dependability of our results. Papers that addressed hybrid optical amplification and WDM-PON design optimization were given preference. Performance-related metrics, such as transmission reach, Q-factor, and bit error rate (BER), were also taken into account. In order to analyze the article's direction, design, and conclusions, paper abstracts would first be checked for relevancy before complete texts were read when relevant.

B. *Literature Review Integration*

In order to learn more about the development, design, and practical implementation of FTTH systems utilizing Wavelength Division Multiplexing Passive Optical Network technology, the research begins with a comprehensive analysis of the literature. The assessment looks for important architectural features that support the growth of high-speed broadband. It then evaluates how important certain parameters—like the

splitting ratio, modulation format, and wave length allocation—are to enhancing the network's scalability and efficiency. The paper covers a wide range of optical amplification techniques, including hybrid, Raman, and Erbium-Doped Fiber Amplifiers (EDFA), and how they affect transmission distances and the integrity of optical data.

The literature review also examined previous comparative studies comparing Coarse Wavelength Division Multiplexing (CWDM) and Dense Wavelength Division Multiplexing (DWDM) techniques, highlighting variations in trade-offs in cost, capacity, and spectrum efficiency. The literature analysis identifies performance indicators for optical signal-to-noise ratio (OSNR), bit error rate (BER), and transmission reach in reliable networks.

The literature evaluation analyzed the studies and identified research gaps in amplification approaches/optimization strategies, signal uniformity, and deployment model efficiency. These findings explain the methodological choices that have shaped project decisions, such as amplifier placement, wavelength grid, and optical power budgeting.

C. *System Planning and Architecture Design*

In the first part of the research, an organized strategy was used to overlay the Wavelength Division Multiplexing Passive Optical Network (WDM-PON) architecture onto the Fiber-to-the-Home (FTTH) architecture. This architecture required to generate high data rates, efficiently use bandwidth at a feed for consumers, and be designed for economic efficiency and scalability. The project proposes a point-to-multipoint (P2MP) topology, with a central office OLT distributing data to many ONUs using passive optical splitters or Arrayed Waveguide Grating (AWG). Each user will be granted a different wavelength (channel) in order to provide dedicated (guaranteed) bandwidth while limiting interference [1].

The setup will make use of several Distributed Feedback (DFB) laser sources, with each laser source modulated to broadcast a specific wavelength using a Dense Wavelength Division Multiplexing (DWDM) multiplexer. The appropriate parameters (wavelength separation, transmission strength, and modulation format) were chosen to preserve the long-distance signal. During the planning phase, practical considerations were made regarding prices, fiber deployment possibilities, and future service scalability. This was backed by previous research [15], which found that WDM-PON architectures enabled flexible bandwidth sharing and better performance across FTTH systems compared to GPON [16].

D. *Performance Simulation and Evaluation*

The fifth stage centered on simulating, assessing, and validating the proposed WDM-PON FTTH architecture with 'OptiSystem' as the simulation tool. The goal was to evaluate system performance across various configurations while considering physical constraints such as transmission distance, amplifier placement, and optical power levels. For performance metrics of the WDM-PON FTTH architecture, we focused on the reliance of Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Quality Factor (Q-factor) to measure the system's dependability and signal quality. Our simulation tests focused on performance indicators such as Bit Error Rate (BER), Signal-to-Noise Ratio (SNR), and Quality Factor (Q factor). The performance of each design was assessed at transmission distances ranging from 10 km to 100 km, with data rates of up to 20 Gbps per channel in a WDM-PON FTTH system. We compared Erbium-Doped Fiber Amplifiers (EDFA) with Raman amplifiers for signal amplification and noise reduction at different rates, both alone and in combination.

We reviewed the simulation results to determine the best balance of capacity, link distance, and cost. When compared to typical GPON models in the literature, the WDM-PON system with hybrid amplification demonstrated better BER values and overall signal quality for long-distance transmission. Ahmad et al. reported similar results. Furthermore, Al-Mashhadani and Reddy (2020 and 2022) found that WDM-PON is appropriate for long-distance FTTH design and that hybrid EDFA-Raman amplification considerably increased optical link performance.

IV. SYSTEM DESIGN AND TESTING

The research presented here provides a complete and objective critique of previous studies on gain and power optimisation in WDM-PON-based FTTH networks. The study evaluates earlier experimental and simulation-based work to identify crucial parameters influencing optical signal strength, transmission reach, and overall network efficiency [19].

The systematic analytical method guarantees a transparent, sequential analysis of how gain adjustment, power budgeting, and amplifier positioning enhance coverage and performance. This comprehensive evaluation allows for recognizing existing technological shortcomings and optimization opportunities that need additional exploration to improve the reliability and scalability of future optical access networks [13]

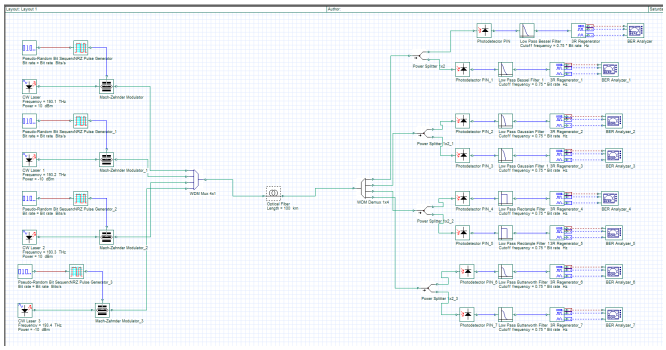


Fig. 1. 4 Channel WDM-PON Model.

The developed system is a 4-channel Wavelength Division Multiplexing Passive Optical Network (WDM-PON) that facilitates effective sharing of bandwidth among several users. Every channel employs a Pseudo-Random Bit Sequence (PRBS) generator, a Non-Return-to-Zero (NRZ) pulse generator, a Continuous Wave (CW) laser, and a Mach-Zehnder Modulator (MZM) to generate optical signals at different wavelengths.

These signals are merged with a WDM multiplexer and sent through an optical fiber connection.

At the receiving end, a WDM demultiplexer divides the wavelengths, and optical splitters send them to various receivers, where the optical signals are transformed back into electrical signals. A Bit Error Rate (BER) analyzer subsequently assesses the quality and effectiveness of the signal. This arrangement demonstrates how several users can effectively share one optical fiber by sending data at different wavelengths with little interference.

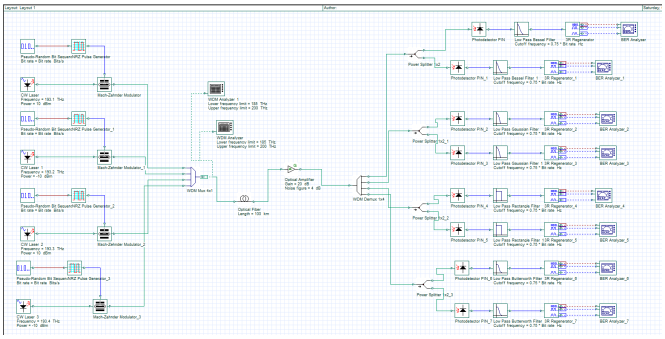


Fig. 2. 4 Channel WDM-PON Model Utilizing Gain and Power Analysis for Improved FTTH Coverage.

The suggested model illustrates a Wavelength Division Multiplexing Passive Optical Network (WDM-PON) intended for Fiber-to-the-Home (FTTH) uses, concentrating on evaluating optical gain and power to improve network coverage. Several Continuous Wave (CW) lasers working at different wavelengths are utilized as light sources, with each being modulated with Non-Return-to-Zero (NRZ) data through the use of Mach-Zehnder Modulators (MZMs).

The modulated signals are then combined via a WDM multiplexer to generate a composite optical signal. To compensate for the fiber attenuation and increase the transmission distance, an optical amplifier is placed behind the multiplexer to amplify the signal power [11]. The amplified signal is transmitted via a fiber optic link, with transmission distances varying from 50 km to 300 km to evaluate system performance under different conditions. On the receiver side, a WDM demultiplexer separates the multiplexed wavelengths, which are then detected by photodiodes, filtered, and regenerated for precise data recovery [18].

TABLE I
SYSTEM PARAMETERS FOR WDM-PON BASED FTTH ARCHITECTURE

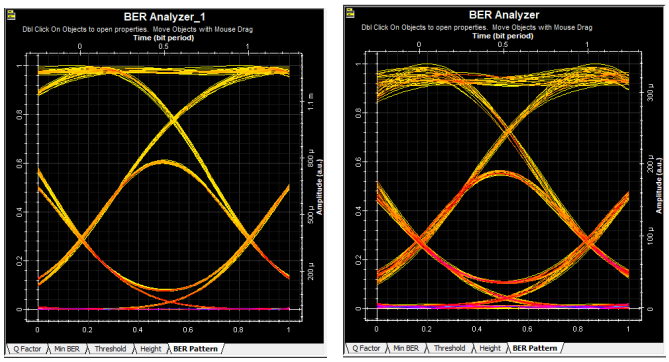
Parameter	**Value**
Wavelength Range	193.1–193.4 THz (1550 nm band)
Number of Channels	4 (WDM Multiplexed)
Modulation Format	NRZ using Mach–Zehnder Modulator
Bit Rate per Channel	10 Gbps
Fiber Type	Single Mode Fiber (SMF)
Fiber Length	100 km
Optical Amplifier Gain	15 dB
Noise Figure of Amplifier	4.5 dB
Optical Splitter Ratio	1×16 per branch
WDM Multiplexer/Demultiplexer	4×1 MUX and 1×4 DEMUX
Photodetector Type	PIN Diode
Photodetector Responsivity	0.9 A/W
Photodetector Dark Current	10 nA
Filter Types Used	Bessel
Regenerator Type	3R (Reshape, Retiming, Regeneration)
Laser Power per Channel	0 dBm
Channel Spacing	0.8 nm (100 GHz)
Simulation Tool	OptiSystem
Performance Metrics	BER, Eye Diagram, Q-Factor

The system performance is analyzed using Bit Error Rate (BER) and eye-illustration measures. The results indicate that effective gain control and power optimization significantly enhance both the content and trustability of FTTH networks employing WDM-PON infrastructure.

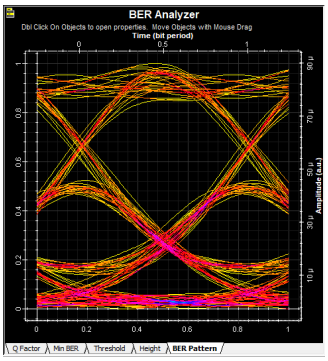
V. RESULTS & DISCUSSIONS

4-Channel WDM-PON:

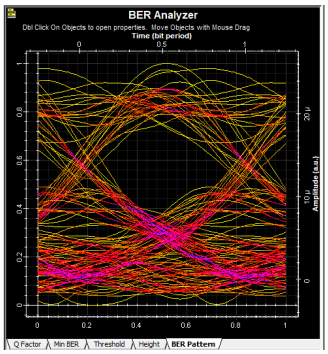
Iterations:



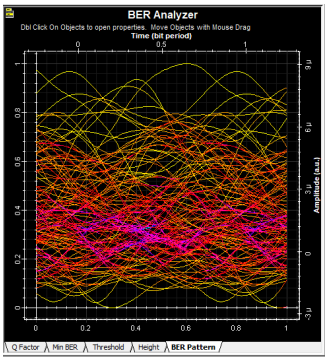
(a) Distance(km) = 30 (b) Distance(km) = 60



(c) Distance(km) = 90



(d) Distance(km) = 120



(e) Distance(km) = 150

Fig. 3. Comparison of WDM-PON architectures Eye Diagrams - measuring different distances.

Data table for 4-Channel WDM-PON:

TABLE II
SIMULATION RESULTS FOR FIBER DISTANCE VS Q FACTOR AND BER

Iteration No.	Distance (km)	Max. Q Factor	Min. BER
1	30	4.50857	2.44E–06
2	60	3.88391	4.14E–05
3	90	4.50715	3.26E–06
4	120	3.38710	3.38E–04
5	150	0.00000	1.00E+00

Graph of the 4-Channel WDM-PON

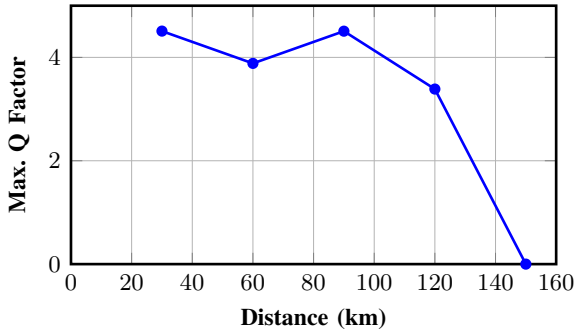


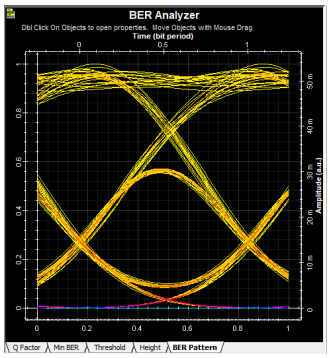
Fig. 4. Graph of the 4-Channel WDM-PON

The simulation results of the 4-Channel WDM-PON system clearly illustrate the performance variation with increasing transmission distance. The eye diagrams obtained at different distances demonstrate how signal quality degrades as fiber length increases due to attenuation and dispersion effects. At shorter distances (30–90 km), the eye openings are wide and distinct, signifying strong signal integrity and minimal intersymbol interference. However, beyond 120 km, the eye becomes narrower and more distorted, indicating a reduction in signal clarity and an increase in noise. At 150 km, the eye diagram almost closes completely, corresponding to a complete signal failure with a BER of 1.00E+00.

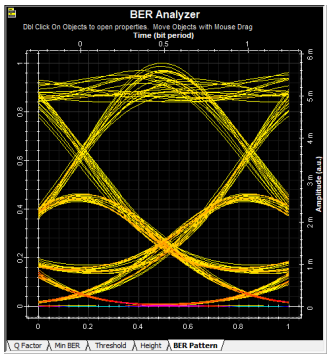
The data table supports these observations quantitatively. The maximum Q-factor initially remains above 4.5 for 30–90 km distances, which ensures acceptable performance with very low BER values (10^{-} – 10^{-} range). As distance increases to 120 km, the Q-factor drops to 3.38 and BER increases to 3.38E-04, revealing signal distortion and degradation. Beyond 150 km, the system performance collapses, showing a Q-factor of 0, which confirms that without amplification, the optical power is insufficient for reliable data transmission over longer distances.

4-Channel WDM-PON Model Utilizing Gain and Power Analysis for Improved FTTH Coverage:

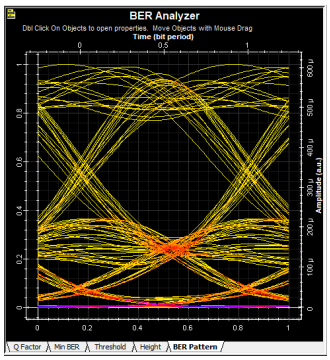
Iterations:



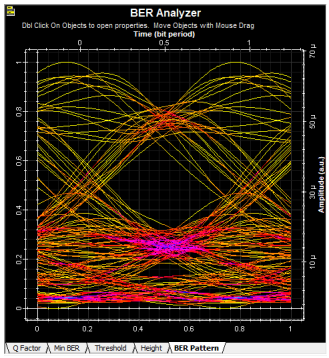
(a) Distance(km) = 30



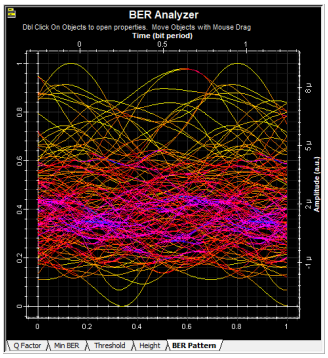
(b) Distance(km) = 60



(c) Distance(km) = 90



(d) Distance(km) = 120



(e) Distance(km) = 150

Fig. 5. Comparison of WDM-PON architectures Eye Diagrams - measuring different distances.

Data table for 4-Channel WDM-PON Model Utilizing Gain and Power Analysis for Improved FTTH Coverage:

TABLE III
SIMULATION RESULTS FOR FIBER DISTANCE VS Q FACTOR AND BER

Iteration No.	Distance (km)	Max. Q Factor	Min. BER
1	50	4.20471	1.05E-05
2	100	4.29962	8.36E-06
3	150	4.19188	1.32E-05
4	200	4.03981	2.55E-05
5	250	0.00000	1.00E+00

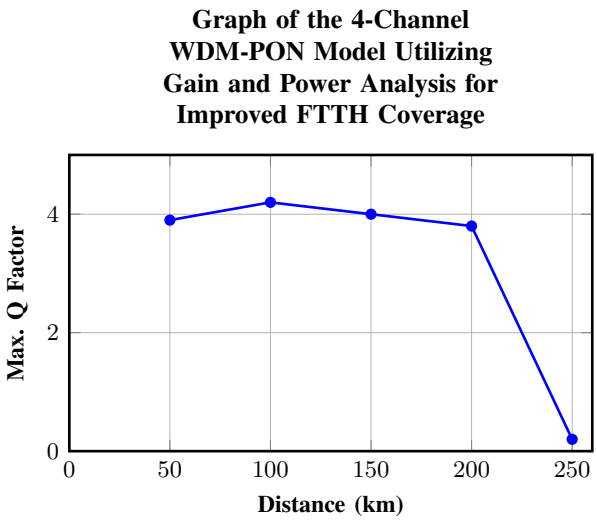


Fig. 6. Graph of the 4-Channel WDM-PON Model Utilizing Gain and Power Analysis for Improved FTTH Coverage

When gain and power optimization are introduced in the improved WDM-PON model, a significant enhancement in performance is observed. The eye diagrams for distances up to 200 km remain wide and symmetrical, reflecting improved signal strength and lower noise levels. The corresponding data table shows stable Q-factors around 4.0–4.3 and BER values as low as 1.05E-05 to 2.55E-05, which fall within the acceptable limit for optical communication. Only beyond 250 km does the system exhibit a complete signal drop, indicating the effective transmission limit for the designed configuration.

The comparison graph between the two models highlights that implementing optical amplification and gain control substantially extends the network's coverage and maintains better performance across longer distances. The optimized system sustains reliable communication up to approximately 200 km, whereas the unamplified design fails beyond 120 km. This clearly demonstrates the crucial role of power balancing and gain adjustment in achieving higher Q-factor stability and lower BER in FTTH networks utilizing WDM-PON architecture.

Moreover, research shows that the optical power budget is crucial for establishing the longest transmission distance and the scalability of the network as a whole. Evaluations through analysis and simulations reveal that precisely fine-tuning the transmitter power, the ratio of optical splitters, and the gain of amplifiers can greatly enhance the area reached by the network while keeping the Bit Error Rate (BER) and Q-factor within acceptable limits [19]. Nonetheless, increasing the gain of the amplifier might lead to amplified spontaneous emission (ASE) noise and nonlinear distortions, highlighting the necessity of achieving a proper balance between gain and loss.

As network infrastructures evolve to higher capacities, the integration of dynamic gain control and adaptive power management becomes imperative. These solutions allow gain levels to be adjusted in real time based on different traffic conditions, fiber losses and environmental factors, thereby increasing reliability and energy efficiency [10]. Additionally, WDM-PON's ability to assign dedicated wavelengths to each user provides better security, traffic isolation, and quality of service than traditional TDM-PON systems but requires precise wavelength stabilization and temperature control mechanisms.

The findings emphasize that the performance of WDM-PON-grounded FTTH systems depends on a holistic design approach combining optic element optimization, intelligent control algorithms, and effective power allocation strategies. As optic access networks continue to expand, unborn exploration should explore AI-driven optimization fabrics, machine literacy-grounded power vaticination models, and energy-apprehensive amplifier designs to achieve lesser scalability, reduced functional costs, and enhanced end-user experience.

VI. CONCLUSION

This study thoroughly examined the effect of optical gain and power distribution on the performance and coverage of Fiber-to-the-Home (FTTH) networks using the Wavelength Division Multiplexing Passive Optical Network (WDM-PON) architecture. The results demonstrate that optimising optical power and amplifier gain has a direct impact on system reach, bandwidth utilisation, and overall signal integrity.

Through thorough power budget evaluation and simulation of network performance across different transmission distances, attenuation factors, and gain settings, it was found that a well-adjusted gain enhances the optical signal sufficiently to counteract splitter and fiber losses while ensuring noise

levels remain within acceptable limits. The use of optical amplifiers like EDFAs (Erbium-Doped Fiber Amplifiers) and SOAs (Semiconductor Optical Amplifiers) demonstrated that the network range can be increased beyond typical passive boundaries without compromising the BER performance

Furthermore, WDM-PON design has shown itself to be a valuable method for boosting FTTH scalability since various wavelengths can be assigned to individual users without interference between channels. This strategy allows for increased data transfer rates, greater spectrum utilization, and enhanced Quality of Service (QoS) when compared to conventional TDM-PON systems. Nevertheless, it also brings about difficulties in channel spacing, wavelength consistency, and power leveling that need to be meticulously handled using accurate gain control methods.

The analysis also highlights that inordinate modification can beget nonlinear deformations and ASE(Amplified robotic Emission) noise, which negatively affect performance. Hence, achieving optimal gain-to-loss balance is pivotal for maintaining signal quality while minimizing energy consumption and outfit costs.

In conclusion, effective management of optical gain and power parameters in WDM-PON-based FTTH networks is essential to expand coverage, ensure stable connectivity, and support the ever-increasing demand for high-speed broadband access. Future developments should focus on dynamic power control, AI-based gain optimization, and intelligent monitoring systems that can adapt to real-time traffic and environmental changes. Such innovations will pave the way for more resilient, energy-efficient and high-capacity optical access networks that can meet the needs of the next generation of smart cities and digital ecosystems.

VII. ACKNOWLEDGEMENT

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